ENN583 Foundations of Robotic Vision

Practice Exam, 2026

QUESTION 1

Which output is produced by the description stage of an image feature pipeline?

(a) The final camera trajectory

(b) A representation of the appearance around each detected feature

(c) A list of camera exposure times only

(d) The three-dimensional position of every scene point

QUESTION 2

An indoor robot observes a floor made from repeating square tiles. What problem is most likely to occur during feature matching?

(a) Several tiles may provide plausible matches for the same detected feature.

(b) Perspective projection is disabled by repeating textures.

(c) Descriptors from all tiles must have different dimensions.

(d) Feature matching becomes unnecessary because the floor is planar.

QUESTION 3

Which observation most directly satisfies the FAST corner criterion presented in the lecture?

(a) A long contiguous arc of circle pixels is darker than the centre by more than the threshold.

(b) Several separated circle pixels have exactly the same value as the centre.

(c) The whole image has a high average brightness.

(d) The candidate has already been matched in another image.

QUESTION 4

During feature matching, the nearest descriptor distances are 35 and 40. The ratio-test threshold is 0.80, and acceptance requires a ratio strictly below the threshold. What is the result?

(a) Retain the match because $35 < 40$.

(b) Retain the match because every ratio below one is accepted.

(c) Reject the match because $35/40 = 0.875$ is not below 0.80.

(d) Reject the match because the ratio is greater than one.

QUESTION 5

SuperGlue receives local features that have already been detected and described, then produces correspondences for geometric pose estimation. What role does SuperGlue play in this pipeline?

(a) It captures the images and calibrates the camera before feature extraction.

(b) It replaces the camera with a wheel-odometry sensor.

(c) It requires every feature to be accepted as a match.

(d) It uses extracted local features to produce correspondences for the geometric back end.

QUESTION 6

Why is LightGlue's ability to stop processing an easy image pair earlier useful for a mobile robot?

(a) It guarantees that every easy pair contains more correct matches.

(b) It removes the need to detect or describe local features.

(c) It makes runtime independent of the amount of computation performed.

(d) It can reduce average latency and computation while using more work for difficult pairs.

QUESTION 7

In the pinhole-camera equation, what is the role of the intrinsic matrix $\mathbf{K}$?

(a) It determines the world-frame position of every object without a pose.

(b) It maps camera-frame coordinates into image coordinates using the camera's internal parameters.

(c) It matches feature descriptors between images.

(d) It stores the camera's frame-to-frame translation history.

QUESTION 8

You are comparing stereo camera systems from two manufacturers. Camera system 2 has a larger physical baseline than system 1. Which trade-off applies to camera system 2?

(a) Higher depth error and a smaller minimum measurable depth in every case

(b) No effect because baseline is absent from the depth equation

(c) Lower depth error, but a larger minimum measurable depth and harder matching for close objects

(d) Elimination of all occlusions between the views

QUESTION 9

A feature matcher currently searches a square area in the second stereo image. What change is justified after the calibrated images have been rectified?

(a) Restrict the search to the corresponding image column.

(b) Accept only a feature at the identical horizontal coordinate.

(c) Restrict the search to the corresponding image row.

(d) Remove all geometric restrictions from the search.

QUESTION 10

A point has depth $Z = 4\,\mathrm{m}$ and appears at image coordinate $u = 270$. The camera parameters are $c_u = 320$ and $f = 500$ pixels. What is its X coordinate in the camera frame?

(a) $-0.40\,\mathrm{m}$

(b) $0.40\,\mathrm{m}$

(c) $-4.0\,\mathrm{m}$

(d) $40\,\mathrm{m}$

QUESTION 11

After a feature is selected in image 1, what geometric restriction does the Fundamental matrix impose on its match in image 2?

(a) The match must have the same descriptor value in every component.

(b) The match must lie on the corresponding epipolar line.

(c) The match must be at the principal point.

(d) The match must always be on the same row, even for unrectified cameras.

QUESTION 12

Why can RANSAC recover a useful geometric model even when the full match set contains outliers?

(a) Every random sample is guaranteed to contain only inliers.

(b) Descriptor distance and geometric error are always identical.

(c) Some random minimal samples may contain only inliers and generate a model supported by many matches.

(d) RANSAC changes the camera intrinsics until all matches agree.

QUESTION 13

A RANSAC inlier threshold is changed from 2 pixels to 5 pixels while the model hypothesis and residuals remain fixed. Which statement is correct?

(a) Every previous inlier becomes an outlier and some previous outliers may become inliers.

(b) Every previous inlier remains an inlier, and some previous outliers may become inliers.

(c) The inlier set must remain exactly unchanged.

(d) The inliers will now produce a better solution than before when refining them with least squares.

QUESTION 14

A monocular 2D–2D visual-odometry system reports translation vector $(0.6, 0, 0.8)$. What additional information is required to report the translation in metres?

(a) A different ordering of the same image correspondences

(b) The rotation matrix

(c) A metric scale estimate

(d) The calibrated Fundamental matrix

QUESTION 15

Using the convention that ${}^{A}\mathbf{T}_{B}$ maps coordinates from frame B to frame A, which composition maps a point from frame C to frame A?

(a) ${}^{A}\mathbf{T}_{C} = {}^{B}\mathbf{T}_{C}\,{}^{A}\mathbf{T}_{B}$

(b) ${}^{A}\mathbf{T}_{C} = ({}^{A}\mathbf{T}_{B})^{-1}\,{}^{B}\mathbf{T}_{C}$

(c) ${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B} + {}^{B}\mathbf{T}_{C}$

(d) ${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C}$

QUESTION 16

RANSAC hypotheses A, B, C, and D are supported by 52, 48, 61, and 55 observations. Which hypothesis wins under a largest-consensus rule?

(a) Hypothesis C

(b) Hypothesis A

(c) Hypothesis B

(d) Hypothesis D

QUESTION 17

A monocular visual-odometry estimate contains rotation and translation direction but no translation distance. Which additional measurement could make the translation metric?

(a) A known scene distance or an external metric sensor measurement

(b) A different permutation of the same 2D correspondences

(c) The number of bits in each descriptor

(d) The image filenames

QUESTION 18

What is the difference between a minimal solver and final pose refinement in a P3P/RANSAC pipeline?

(a) The minimal solver creates hypotheses from three correspondences; refinement uses a larger inlier set for accuracy.

(b) Refinement must discard all correspondences used by RANSAC.

(c) The minimal solver estimates only image colour, while refinement estimates depth.

(d) Both stages must use exactly three correspondences.

QUESTION 19

During PnP with RANSAC, a candidate pose projects a landmark to pixel $(100, 80)$ and the observed pixel is $(97, 84)$. The residual is observed minus predicted and the inlier threshold is 6 pixels. How is the match classified?

(a) Residual $(3, -4)$, error 7 pixels, outlier

(b) Residual $(-3, 4)$, error 25 pixels, outlier

(c) Residual $(197, 164)$, error 5 pixels, inlier

(d) Residual $(-3, 4)$, error 5 pixels, inlier

QUESTION 20

What complementary roles do RANSAC and iterative PnP play?

(a) Iterative PnP detects image corners; RANSAC calculates descriptors.

(b) RANSAC provides robust hypothesis selection; iterative PnP improves the selected pose using its inliers.

(c) RANSAC supplies metric camera intrinsics; iterative PnP changes the image resolution.

(d) Both methods merely reorder the same matches without estimating a pose.

QUESTION 21

If the probability that one sampled correspondence is an inlier is w, then an s-point sample is entirely inlier with probability w^s. What happens as s increases while $0 < w < 1$ remains fixed?

(a) The all-inlier probability decreases exponentially, so more RANSAC trials are required.

(b) The all-inlier probability increases linearly.

(c) The all-inlier probability remains equal to w.

(d) The required RANSAC trials necessarily fall to one.

QUESTION 22

An accumulated trajectory predicts that the relative motion from keyframe 2 to keyframe 8 is T_{chain}. A separate estimate obtained by directly predicting the relative pose between these keyframes estimates T_{direct}, and the two are slightly different. What should a pose graph do with these measurements?

(a) Force both measurements to become numerically identical before constructing the graph.

(b) Discard all consecutive measurements because a non-consecutive measurement is always exact.

(c) Treat both transforms as absolute camera poses in the world frame.

(d) Adjust the pose variables to find a configuration that balances the weighted measurement residuals.

QUESTION 23

In a factor graph,Factor A and Factor B have the same residual. Factor A has standard deviation 0.5, while Factor B has standard deviation 1.0. For a squared normalised residual, which statement is correct?

(a) Factor A contributes twice as much as Factor B.

(b) Factor B contributes four times as much as Factor A.

(c) Factor A contributes four times as much as Factor B.

(d) Their contributions are equal.

QUESTION 24

Which action most directly reduces the risk that a local pose-graph optimiser converges to an implausible solution?

(a) Set every pose and rotation matrix entry to zero.

(b) Remove all measurements that agree with the initial trajectory.

(c) Provide a physically plausible initial trajectory close to the expected solution.

(d) Assume that robust loss functions make initialisation irrelevant.

QUESTION 25

Why can two posterior distributions with the same MAP state imply different decisions for a robot?

(a) A MAP state contains the complete covariance of both distributions.

(b) Their probability mass may have different spread or shape even though their maxima occur at the same state.

(c) Equal MAP states require both posterior distributions to be identical.

(d) The MAP operation changes the robot's coordinate frame.

QUESTION 26

A network must assign one of six terrain classes to every pixel. Which combination is aligned with the task?

(a) One scalar logit, angular regression loss, and mean angle error

(b) Four box coordinates, IoU box loss, and detection average precision

(c) One global embedding, contrastive retrieval loss, and recall at 5

(d) Per-pixel class logits, cross-entropy loss, and mean mask IoU

QUESTION 27

A collision detector stops a robot when its hazard score exceeds a threshold. Which threshold change generally reduces missed hazards at the cost of more unnecessary stops?

(a) Raise the threshold.

(b) Keep the threshold fixed and multiply all reported scores by zero.

(c) Remove the score from the stopping decision.

(d) Lower the threshold.

QUESTION 28

A collision prediction model has high average accuracy, but it often misses a rare safety-critical object and its latency causes stale predictions while the robot is moving. What is the best conclusion?

(a) High average accuracy proves that rare-class recall and real-time behaviour are acceptable.

(b) Latency cannot affect robot behaviour if the model's image labels are accurate.

(c) The rare failures should be removed from evaluation because they reduce the average.

(d) Aggregate accuracy alone is insufficient; class-specific hazards, temporal behaviour, and latency must also be evaluated.

QUESTION 29

Which conclusion is justified by photorealistic novel views from a 3D Gaussian-splat representation?

(a) The representation models appearance well for the tested views, but physical occupancy still requires validation.

(b) Every Gaussian corresponds exactly to a solid object with known metric boundaries.

(c) Every transparent or reflective surface is guaranteed to be reconstructed correctly.

(d) Every volume without an opaque Gaussian is guaranteed collision-free.

QUESTION 30

A learned monocular model outputs metric depth, but tests show large errors on transparent doors in the deployment building. Is metric output alone sufficient for collision avoidance?

(a) Yes; metric units guarantee correct depth at every pixel.

(b) Yes; transparent regions can be treated as free whenever the network reports the background.

(c) No; metric depth cannot be back-projected with camera intrinsics.

(d) No; physical units address scale, but the transparent-surface geometry and coordinate alignment still require validation.

QUESTION 31

An object detection model produces two bounding boxes. Box A has area 60 pixels^2, Box B has area 50 pixels^2, and their intersection has area 30 pixels^2. What is their Intersection over Union?

(a) 0.273

(b) 0.375

(c) 0.50

(d) 0.60

QUESTION 32

What is the main difference between average precision and precision measured at one deployed confidence threshold?

(a) Average precision measures only object size, while deployed precision measures class frequency.

(b) Average precision is recall at one threshold with a different name.

(c) The two measures must always be numerically identical.

(d) Average precision summarises performance across thresholds, while the deployed precision describes one operating point.

QUESTION 33

A segmentation model predicts background everywhere and achieves high pixel accuracy because obstacles are rare. What is the most direct training problem?

(a) The unweighted objective is dominated by the common background class.

(b) The model has too many obstacle pixels in its labels.

(c) Semantic segmentation cannot represent obstacles.

(d) High background accuracy proves obstacle recall is high.

QUESTION 34

Which expression defines Intersection over Union for a predicted mask P and ground-truth mask G?

(a) $|P \cap G|/|G|$

(b) $|P \cap G|/|P \cup G|$

(c) $|P \cap G|/|P|$

(d) $|P \cap G|/|\text{all image pixels}|$

QUESTION 35

A unit-length query descriptor is $q = (0.8, 0.6)$. Database descriptors $a = (0.8, 0.6)$ and $b = (-0.8, -0.6)$ are also unit length. What are their cosine similarities with q?

(a) Both similarities are 1 because all descriptors are unit length

(b) Both similarities are 0 because the components cancel

(c) $q^T a = 1$ and $q^T b = -1$

(d) $q^T a = -1$ and $q^T b = 1$

QUESTION 36

For five place-recognition queries, the first correct database match appears at ranks 2, 5, 1, and 6; no correct match is returned for the fifth query. What is recall@2?

(a) $1/5 = 0.20$

(b) $2/5 = 0.40$

(c) $3/5 = 0.60$

(d) $4/5 = 0.80$

QUESTION 37

A place-recognition system must scale to a large database and avoid false loop closures in repeated corridors. Which design best balances these needs?

(a) Geometrically compare every local feature with the entire database and accept the pair with the most raw matches.

(b) Use only the top global similarity and assume repeated corridors share the same pose.

(c) Use compact global descriptors for candidate search, then rerank using local spatial evidence and accept only geometrically consistent candidates.

(d) Store one average corridor descriptor and accept it for every query.

QUESTION 38

To create a reliability diagram, an engineer groups a classifier's validation predictions into bins according to their confidence scores. A confidence bin contains 10 predictions with mean confidence 0.60. Eight are correct. Which statement is correct?

(a) Accuracy is 0.80, so the bin is overconfident by 0.20.

(b) Accuracy is 0.60, so the bin is perfectly calibrated.

(c) Accuracy is 0.80, so the bin is underconfident by 0.20.

(d) Accuracy is 0.20, so the bin is overconfident by 0.40.

QUESTION 39

A computer vision system processes 50 inputs, makes 40 predictions, and abstains on 10. Four predictions are wrong. What are coverage and selective risk?

(a) Coverage 0.80 and selective risk 0.10

(b) Coverage 0.80 and selective risk 0.08

(c) Coverage 0.20 and selective risk 0.10

(d) Coverage 0.20 and selective risk 0.08

QUESTION 40

Why should a detector's operating threshold not be chosen by repeatedly checking the final test set?

(a) Thresholds can be selected only before any model is trained.

(b) The test set would influence the selected system and no longer provide an independent final estimate.

(c) A test set cannot contain confidence scores.

(d) Repeated test use automatically changes the ground-truth labels.

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